Pulkit Verma

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Research Interests

AI Safety, AI Planning, Interpretability, Symbolic Model Learning, Analysis of Abstractions, and Robotics.

Education

Ph.D. Computer Science 2018-2024	**Arizona State University**, School of Computing and Augmented Intelligence Thesis: *Data-Efficient Paradigms for Personalized Assessment of Taskable AI Systems* Advisor: Siddharth Srivastava
M.Tech. Computer Science 2013-2015	**Indian Institute of Technology Guwahati**, Department of CSE Thesis: *Resource Usage Analysis for Speech Recognition Techniques* Advisor: Pradip K. Das
B.Engg. Information Tech. 2007-2011	**Shri Vaishnav Institute of Technology & Science, Indore**, Department of IT Capstone: Smart Analyzer - Predict and Cache Webpages based on Usage Advisor: Anand S. Rajawat

Research and Professional Experience

Sep 2024 – (present)	**Postdoctoral Research Associate**, CSAIL, Massachusetts Institute of Technology. Advisor: Julie Shah. *Research Areas:* AI Interpretability, Model Learning, Black-Box Assessment
Aug 2018 – Aug 2024	**Graduate Research Associate**, AAIR Lab, ASU. Advisor: Siddharth Srivastava. *Research Areas:* Agent Assessment, Model Learning, Automated Planning
May 2023 – Aug 2023	**Research Scientist Intern**, Meta AI. Mentors: Rohan Chitnis, Alborz Geramifard, Nitin Kamra. *Research Areas:* Automated Planning, Curriculum Personalisation
Jul 2015 – Jul 2018	**Member of Technical Staff**, NetApp Inc. *Predicted workloads* for storage volumes using I/O patterns for All Flash systems.
Aug 2013 – May 2015	**Research Assistant**, IIT Guwahati. Mentors: Pradip K. Das, Shivashankar B. Nair, Shashi Shekhar Jha. *Research Areas:* Speech Processing, Mobile Robotics
Mar 2012 – Aug 2013	**Assistant Systems Engineer,** Tata Consultancy Services Pvt. Ltd. *Developed iOS applications* for various B2C companies.

Teaching Experience

Teaching Assistant	CSE 571 Artificial Intelligence	Spring 2022
Arizona State University	CSE 571 Artificial Intelligence	Fall 2019
Session Lead	CSE 571 Artificial Intelligence	Spring 2023
Arizona State University	CSE 471 Introduction to Artificial Intelligence	Fall 2022
	CSE 574 Planning and Learning Methods in Artificial Intelligence	Spring 2021
	CSE 471 Introduction to Artificial Intelligence	Fall 2020

Teaching Experience (continued)

Teaching Assistant IIT Guwahati	CS 243 Software Engineering Lab	Spring 2015
	CS 566 Speech Processing	Fall 2014
	CS 462 Computer Graphics Lab	Spring 2014
	CS 513 Programming Lab	Fall 2013

Honors and Awards

AAAI/ACM SIGAI Innovative AI Education Award 2025 — January 2025
Awarded to 4 people at AAAI 2025's EAAI Symposium

Graduate College Completion Fellowship, ASU Graduate College — June 2023
Highly competitive (1 per program per year) with tuition, stipend, and insurance for 1 year

Best Demo Award, AAMAS 2022 — May 2022
Awarded to one system demonstration at the AAMAS 2022 conference

Outstanding Reviewer, CoRL 2024 LEAP Workshop — November 2024
Spring 2024 GPSA Jumpstart Research Grant, ASU GPSA — January 2024
Fall 2023 GPSA Outstanding Research Award, ASU GPSA — October 2023
Fall 2023 GPSA Outstanding Mentor Award, ASU GPSA — October 2023
University (Engineering) Graduate Fellowship, ASU Engineering and Graduate College — March 2023
SCAI Doctoral Fellowship, School of Computing and Augmented Intelligence, ASU — March 2023
Spring 2022 GPSA Teaching Excellence Award, ASU GPSA — April 2022
SCAI Doctoral Fellowship, School of Computing and Augmented Intelligence, ASU — March 2022
Fulton Schools Experiential Learning Grant, ASU Engineering — October 2018
Post Graduation Fellowship, All India Council for Technical Education, Govt. of India — 2013-2015
TCS ILP Kudos Award, Tata Consultancy Services Ltd. (Best Trainee in the Batch) — May 2012

Talks

Learning Interpretable Models for Autonomous Assessment of Taskable AI Systems
Department of Computer Science and Engineering, IIT Jodhpur — January 2025
Department of Computer Science and Engineering, IIT Madras — January 2025
School of Data Science and AI, IIT Palakkad — January 2025
School of Data Science and AI, IIT Roorkee — December 2024
School of AI and Data Engineering, IIT Ropar — October 2024
Brown Robotics Talks, BigAI, Brown University — October 2024
Department of Computer Science and Engineering, IIT Guwahati — August 2024
Interactive Robotics Group, Massachusetts Institute of Technology — May 2024
ICAROS Lab, University of Southern California — May 2024

Data-Efficient Paradigms for Personalized Assessment of Taskable AI Systems
CHAI Hands-On, University of California Berkeley — April 2024
Intelligent Robot Learning Lab, University of Alberta — February 2024

Discovering User-Interpretable Capabilities of Black-Box AI Agents
CHAI - 6th Annual Workshop — June 2022

Publications

* denotes equal contribution

Conference Proceedings and Journals

[C1] Rushang Karia*, Jayesh Nagpal*, Daksh Dobhal*, **Pulkit Verma**, Rashmeet K Nayyar, Naman Shah, and Siddharth Srivastava. "Using Explainable AI and Hierarchical Planning for Outreach with Robots". In *39th AAAI Conference on Artificial Intelligence (EAAI Symposium Track)*, 2025. (Accepted) — CORE Ranking: A*

[C2] Rushang Karia*, **Pulkit Verma***, Alberto Speranzon, and Siddharth Srivastava. "Epistemic Exploration for Generalizable Planning and Learning in Non-Stationary Settings". In *34th International Conference on Automated Planning and Scheduling*, 2024. CORE Ranking: A*

[C3] **Pulkit Verma**, Rushang Karia, and Siddharth Srivastava. "Autonomous Capability Assessment of Sequential Decision-Making Systems in Stochastic Settings". In *37th Conference on Advances in Neural Information Processing Systems*, 2023. CORE Ranking: A*

[C4] **Pulkit Verma**, Shashank Rao Marpally, and Siddharth Srivastava. "Discovering User-Interpretable Capabilities of Black-Box Planning Agents". In *19th International Conference on Principles of Knowledge Representation and Reasoning*, 2022. CORE Ranking: A*

[C5] Naman Shah*, **Pulkit Verma***, Trevor Angle, and Siddharth Srivastava. "JEDAI: A System for Skill-Aligned Explainable Robot Planning". In *21st International Conference on Autonomous Agents and MultiAgent Systems (Demonstration Track)*, 2022. ***[Best Demo Award]***. CORE Ranking: A*

[C6] Rashmeet Kaur Nayyar*, **Pulkit Verma***, and Siddharth Srivastava. "Differential Assessment of Black-Box AI Agents". In *36th AAAI Conference on Artificial Intelligence*, 2022. CORE Ranking: A*

[C7] Yizhong Wang, Swaroop Mishra, Pegah Alipoormolabashi, Yeganeh Kordi, Amirreza Mirzaei, Anjana Arunkumar, Arjun Ashok, **Pulkit Verma**, et al. "Super-NaturalInstructions: Generalization via Declarative Instructions on 1600+ Tasks". In *2022 Conference on Empirical Methods in Natural Language Processing*, 2022. CORE Ranking: A

[C8] **Pulkit Verma**, Shashank R Marpally, and Siddharth Srivastava. "Asking the Right Questions: Learning Interpretable Action Models Through Query Answering". In *35th AAAI Conference on Artificial Intelligence*, 2021. Ranking: A*

[C9] **Pulkit Verma**, and Pradip K. Das. "A Comparative Study of Resource Usage for Speaker Recognition Techniques". In *International Conference on Signal Processing and Communication*, 2016.

[C10] **Pulkit Verma**, and Pradip K. Das. "i-Vectors in Speech Processing Applications: A Survey". In *International Journal of Speech Technology*, 18(4), pp.529-546, 2015. Impact Factor: 2.74

[C11] Mayank Gupta, **Pulkit Verma**, Tuhin Bhattacharya, and Pradip K. Das. "A Mobile Agents based Distributed Speech Recognition Engine for Controlling Multiple Robots". In *International Conference on Advances In Robotics*, 2015.

[C12] **Pulkit Verma**, Mayank Gupta, Tuhin Bhattacharya, and Pradip K. Das. "Improving Services using Mobile Agents-based IoT in a Smart City". In *International Conference on Contemporary Computing and Informatics*, 2014.

Peer-Reviewed Workshop and Symposia Papers (Non-archival)

[W1] Dillon Ze Chen, **Pulkit Verma**, Siddharth Srivastava, Michael Katz, and Sylvie Thiébaux. "AI Planning: A Primer and Survey (Preliminary Report)'. In *AAAI Workshop on Bridging the Gap Between AI Planning and RL*, 2025. (Accepted)

[W2] **Pulkit Verma**. "Bridging the Language Divide: Generative AI's Promise for Global Education". In *15th Symposium on Educational Advances in Artificial Intelligence (Blue Sky Ideas Track)*, 2025. (Accepted) ***[AAAI/ACM SIGAI Innovative AI Education Award]***

[W3] **Pulkit Verma** and Siddharth Srivastava."User-Aligned Autonomous Capability Assessment of Black-Box AI Systems". In *AAAI Spring Symposium on User-Aligned Assessment of Adaptive AI Systems*, 2024.

[W4] Rushang Karia, Daksh Dobhal, Daniel Bramblett, **Pulkit Verma**, and Siddharth Srivastava. "Can LLMs translate SATisfactorily? Assessing LLM Capabilities in Generating Formal Specifications". In *AAAI Spring Symposium on User-Aligned Assessment of Adaptive AI Systems*, 2024.

[W5] **Pulkit Verma***, Rushang Karia*, Gaurav Vipat, Anmol Gupta, and Siddharth Srivastava. "Learning AI-System Capabilities under Stochasticity". In *NeurIPS Workshop on Generalization in Planning*, 2023.

[W6] Rushang Karia*, **Pulkit Verma***, Gaurav Vipat, and Siddharth Srivastava. "Epistemically Guided Exploration for Transferable Stochastic Planning in Non-stationary Relational Settings". In *NeurIPS Workshop on Generalization in Planning*, 2023.

[W7] **Pulkit Verma** and Siddharth Srivastava. "Learning Causal Models of Autonomous Agents using Interventions". In *IJCAI Workshop on Generalization in Planning*, 2021.

[W8] **Pulkit Verma**, Shashank Rao Marpally, and Siddharth Srivastava. "Learning User-Interpretable Descriptions of Black-Box AI System Capabilities". In *ICAPS Workshop on Knowledge Engineering for Planning & Scheduling*, 2021.

[W9] **Pulkit Verma** and Siddharth Srivastava. "Learning Interpretable Models for Black-Box Agents". In *ICML Workshop on Human In the Loop Learning*, 2020.

[W10] **Pulkit Verma**, and Siddharth Srivastava. "Learning Generalized Models by Interrogating Black-Box Autonomous Agents". In *AAAI Workshop on Generalization in Planning*, 2020.

[W11] Alok Shankar Mysore, Vikas S. Yaligar, Imanol Arrieta Ibarra, Camelia Simoiu, Sharad Goel, Ramesh Arvind, Chiraag Sumanth, Arvind Srikantan, Bhargav HS, Mayank Pahadia, Tushar Dobha, Atif Ahmed, Mani Shankar, **Pulkit Verma**, et al. "Investigating the 'Wisdom of Crowds' at Scale.". In *Adjunct Proceedings of the 28th Annual ACM Symposium on User Interface Software and Technology (Poster)*, 2015.

Preprints/Under Review

[P1] Rushang Karia, Daniel Bramblett, Daksh Dobhal, **Pulkit Verma**, and Siddharth Srivastava. "∀uto∃val: Autonomous Assessment of LLMs in Formal Synthesis and Interpretation Tasks". 2024.

[P2] Naman Shah, Jayesh Nagpal, **Pulkit Verma**, and Siddharth Srivastava. "From Reals to Logic and Back: Inventing Symbolic Vocabularies, Actions, and Models for Planning from Raw Data". 2024.

[P3] Mehdi Dadvar, Keyvan Majd, Elena Oikonomou, **Pulkit Verma**, Georgios Fainekos, and Siddharth Srivastava. "Joint Communication and Motion Planning for Cobots in Real-World Contexts". 2024.

Doctoral Consortium

[D1] **Pulkit Verma**. "Data Efficient Paradigms for Personalized Assessment of Black-Box Taskable AI Systems". In *38th AAAI Conference on Artificial Intelligence*, 2024.
Mentor: Sheila McIlraith

[D2] **Pulkit Verma**. "Sample Efficient Paradigms for Personalized Assessment of Taskable AI Systems". In *32nd International Joint Conference on Artificial Intelligence*, 2023.
Mentor: Peter Stone

[D3] **Pulkit Verma**. "Data Efficient Paradigms for Personalized Assessment of Taskable AI Systems – Dissertation Abstract". In *32nd International Conference on Automated Planning and Scheduling*, 2022.
Mentor: Eva Onaindia

[D4] **Pulkit Verma**. "Data Efficient Paradigms for Personalized Assessment of Taskable AI Systems". In *19th International Conference on Principles of Knowledge Representation and Reasoning*, 2022.
Mentor: Hector Geffner

[D5] **Pulkit Verma**. "Data Efficient Algorithms and Interpretability Requirements for Personalized Assessment of Taskable AI Systems". In *30th International Joint Conference on Artificial Intelligence*, 2021.
Mentor: Leslie Pack Kaelbling

Theses

[T1] **Pulkit Verma**. "Data-Efficient Paradigms for Personalized Assessment of Taskable AI Systems". (PhD Dissertation). Arizona State University, Tempe, AZ.

[T2] **Pulkit Verma**. "Resource Usage Analysis for Speech Recognition Techniques". (Masters Thesis). Indian Institute of Technology Guwahati, Guwahati, India.

Technical Reports

[TR1] **Pulkit Verma** and Siddharth Srivastava. “Learning Causally Accurate Models for Autonomous Assessment of Deterministic Black-Box Agents”. Technical Report TR-ASUSCAI-2024-001, School of Computing and Augmented Intelligence, Arizona State University, 2024.

Intellectual Property

[IP1] Siddharth Srivastava and **Pulkit Verma**. “ Systems and Methods for Independent Audit and Assessment Framework for AI Systems”. US Patent Application 63/597,259.

Academic Service

Conference Tutorials

AAAI 2025 Tutorial on User-Driven Capability Assessment of Taskable AI Systems 2025
Pulkit Verma and Siddharth Srivastava

Workshop Leadership

AAAI 2024 Spring Symposium on User-Aligned Assessment of Adaptive AI Systems 2023-2024
Co-organizers: Rohan Chitnis, Georgios Fainekos, Hazem Torfah, Siddharth Srivastava

NeurIPS 2023 Workshop on Generalization in Planning (GenPlan’23) 2023
Co-organizers: Siddharth Srivastava, Aviv Tamar, Felipe W. Trevizan

IJCAI 2022 Workshop on Generalization in Planning (GenPlan’22) 2022
Co-organizers: Yuqian Jiang, Rushang Karia, Jendrik Seipp

Journal Reviewer

Artificial Intelligence Journal (AIJ)	2024-(present)
Neural Computing and Applications (NCAA)	2024-(present)
Journal of Artificial Intelligence Research (JAIR)	2023-(present)
IEEE Robotics and Automation Letters (RA-L)	2021-(present)

Conference Program Committee/Reviewer

AAAI Conference on Artificial Intelligence (AAAI)	2023, 2024, 2025
International Conference on Autonomous Agents and Multi-Agent Systems (AAMAS)	2023, 2024, 2025
International Conference on Automated Planning and Scheduling (ICAPS)	2022, 2024, 2025
International Conference on Learning Representations (ICLR)	2024, 2025
Robotics: Science and Systems Conference (R:SS)	2024, 2025
Advances in Neural Information Processing System (NeurIPS)	2023, 2024
International Joint Conference on Artificial Intelligence (IJCAI)	2023, 2024
International Conference on Artificial Intelligence and Statistics (AISTATS)	2025
IEEE Pacific Visualization Conference (PacificVis)	2025

Academic Service (continued)

Workshop Program Committee/Reviewer

ICLR Workshop on AI Verification in the Wild (VerifAI)	2025
AAAI Workshop on Planning in the Era of LLMs	2025
IJCAI/AAAI Workshop on Generalization in Planning (GenPlan)	2021, 2025
AAAI/IJCAI Workshop on Bridging the Gap Between AI Planning and RL (PRL)	2023, 2025
CoRL Workshop on Learning Effective Abstractions for Planning (LEAP)	2023, 2024
NeurIPS Workshop on Explainable/Interpretable AI in Action (AIA)	2023, 2024
ICAPS Workshop on Human-Aware and Explainable Planning (HAXP)	2023
CVPR Workshop on Open-Domain Reasoning Under Multi-Modal Settings (ODRUM)	2023
ICAPS Workshop on Explainable AI Planning (XAIP)	2021, 2022

Miscellaneous Academic Service

Co-author, ONR grant on Adaptive Training and Education for Adaptive AI Systems	2023
Member, (Inaugural) PhD Council, School of Computing and AI, ASU	2023-2024
Award and Grant Reviewer, Graduate and Professional Service Association (GPSA), ASU	2020-2022

Volunteer

International Joint Conference on Artificial Intelligence (IJCAI)	2021, 2022
International Conference on Principles of Knowledge Representation & Reasoning (KR)	2018, 2022
International Conference on Learning Representations (ICLR)	2021
International Conference on Automated Planning and Scheduling (ICAPS)	2021
Southwest Robotics Symposium	2019
Workshop on GPU Programming and Applications, IITG	2014

Mentoring

PhD Mentees

Josh Rountree	Massachusetts Institute of Technology	2024-(present)
Ngoc La	Massachusetts Institute of Technology	2024-(present)
Daniel Bramblett	Arizona State University	2024-(present)

Masters Mentees

Samir Wadhwania	Massachusetts Institute of Technology	2024-(present)
Jayesh Nagpal	Arizona State University	2022-2024
Daksh Dobhal	Arizona State University	2023-2024
Gaurav Vipat	Arizona State University	2023-2024
Trevor Angle	Arizona State University	2021-2022
Shashank Rao Marpally	Arizona State University	2020-2021

Undergrad Mentees

Dylan Fulop	Arizona State University	2022-2023
Kyle J. Atkinson	Arizona State University	2021-2023
Masih Tamsoy	Indian Institute of Technology Guwahati	2014-2015

Outreach Activities

Doctoral Mentor, School of Computing and Augmented Intelligence, ASU — 2023-2024
Mentored junior PhD students and helped them navigate through various issues

JEDAI Team Member, AAIR Lab, Arizona State University — 2022-2023
Visited High Schools in Tempe, AZ and taught basics of robot planning using JEDAI software

Volunteer, Child Rights and You — 2016-2018
Taught and mentored primary school children in Koramangala PAG, Bengaluru

Student Volunteer, Stanford Scholar Initiative — 2016-2018
Translated and recorded Python tutorial and research talks in Hindi

Other Affiliations

Member, International Association for Safe and Ethical AI (IASEAI)	2025-(present)
Member, Association for the Advancement of Artificial Intelligence (AAAI)	2019-(present)
Graduate Student, Center for Human, AI, and Robot Teaming, ASU	2021-2024
Affiliated Graduate Student, Center for Human-Compatible AI, UCB	2020-2024
Graduate Student, Autonomous Agents and Intelligent Robots Lab, ASU	2018-2024
Participant, 3rd Summer School on Cognitive Robotics, USC	July 2019
Online Contributor, Stanford Scholar Initiative	2016
Graduate Student, Robotics Lab, IIT Guwahati	2014-2015

References Available on Request